

Problem Set 1

Before you begin, please read the following instructions carefully:

- Start early.
- Your submission should include:
 - One document with the answers
 - The code used to answer the questions that require the **R**.
 - Submit your work using the same folder structure you have been using for the lab assignments.
- Late submission will not be accepted. Submission date: before Lab on Sept 24th.
- For the questions which require the use of **R**, make sure you include fully annotated code as part of your answers. For mathematical questions, you should include your detailed derivations.

Questions:

1. Consider two $N \times N$ matrices, A and B . Show that computing their product, $C = AB$, has computational complexity $O(N^3)$.
2. Let $X_1, \dots, X_n$ be i.i.d random variables with $E(X_i) = \mu$. Consider the sample mean $T_n = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. For each i , define the leave-one-out (loo) sample mean

$$T_{-i} = \frac{1}{n-1} \sum_{j \neq i} X_j$$

and let

$$\bar{T}_{(-)} = \frac{1}{n} \sum_{i=1}^n T_{-i}$$

(a) Show that

$$T_{-i} = \frac{n\bar{X} - X_i}{n-1}$$

(b) Show that the average of the leave-one-out estimates equals the original sample mean i.e.,

$$\bar{T}_{(-)} = \bar{X}$$

(c) The estimate of bias is defined as:

$$\widehat{\text{Bias}}_{\text{loo}} = (n-1) (\bar{T}_{(-)} - T_n)$$

In this scenario, calculate this bias estimate term.

(d) Calculate the the following bias-corrected estimator

$$T_{\text{loo}} = T_n - \widehat{\text{Bias}}_{\text{loo}}$$

Explain why this leave-one-out procedure does not change the sample mean.

3. Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from the uniform distribution on $[0, 1]$. Define the sample minimum as $T_n = \min\{X_1, \dots, X_n\}$

- (a) Show that $nT_n \xrightarrow{d} \text{Exp}(1)$, where $\text{Exp}(1)$ denotes the exponential distribution with rate parameter 1.
 - (b) Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from the original sample, and define $T_n^* = \min\{X_1^*, \dots, X_n^*\}$. Show that the probability that T_n is not selected in the bootstrap sample converges to e^{-1} as $n \rightarrow \infty$.
 - (c) Using the result from part 3.2, find the limiting value of $\Pr(T_n^* = T_n)$. What does this imply about the bootstrap distribution of T_n^* relative to the sampling distribution of T_n ? Is the nonparametric bootstrap valid in this setting?
4. Muriel Bristol claimed that she could distinguish between cups in which tea was poured first and cups in which milk was poured first.

Ronald Fisher presents the Muriel with eight cups. In four cups, the tea is poured first; in the other four, the milk is poured first. The order of the eight cups is randomized. The participant knows that there are four cups of each type and must classify exactly four as *tea first* and four as *milk first*. She classified all eight cups correctly.

Throughout the question, use the number of correctly classified cups as the test statistic. Conduct a one-sided test, where larger values provide stronger evidence against the null hypothesis.

- (a) State the sharp null hypothesis using potential-outcomes notation.
- (b) Under the null hypothesis, how many equally likely assignments are possible?
- (c) Enumerate all possible assignments and calculate the number of correctly classified cups under each assignment.
- (d) Calculate the exact one-sided permutation p -value:

$$p = \Pr(S \geq S_{\text{obs}} \mid H_0)$$

where S is the number of correctly classified cups and $S_{\text{obs}} = 8$.

- (e) Based on the resulting p -value, would you reject the null hypothesis at the 5% significance level? Briefly explain.
5. Now suppose the experiment in the previous question is repeated using 100 cups. Tea is poured first in 50 cups, and milk is poured first in the remaining 50. A new participant must classify exactly 50 cups as *tea first* and 50 as *milk first*. She correctly classifies 67 of the 100 cups.

In this setting, enumerating the complete permutation distribution is impossible. Therefore:

- (a) Write R code that uses Monte Carlo simulation to approximate the randomization distribution under the sharp null hypothesis. Use at least $B = 100,000$ random assignments.
- (b) For each simulated assignment, calculate the number of correctly classified cups.
- (c) Plot the simulated randomization distribution and indicate the observed statistic i.e., $S_{\text{obs}} = 67$ with a vertical line.
- (d) Estimate the one-sided permutation p -value.
- (e) Based on the estimated p -value, would you reject the null hypothesis at the 5% significance level? Explain what your conclusion means in the context of the experiment.